DUDEKULA NAGAJYOTHI

COMET FWC 068

MATHEMATICS QUESTION PAPER

Set 1 (English Version Only)

General Instructions

1. This question paper contains multiple sections.
2. All questions are compulsory.
3. Read all questions carefully before answering.
4. Use proper mathematical notation wherever required.

Section A

Q1. If

$$A = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix},$$

show that

$$A^T A = I.$$

Q2. If

$$|A| = 5,$$

find the value of

$$|3A|.$$

Q3. Find the value of k such that the system of equations has a unique solution:

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$2x + 3y + kz = 15$$

Q4. Write the sum of intercepts cut off by the plane

$$\vec{r} \cdot (2\hat{i} + 5\hat{j} + \hat{k}) = 10$$

on the coordinate axes.

Q5. If

$$\vec{a} = 4\hat{i} + \hat{j} + \hat{k}$$

and

$$\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k},$$

find a unit vector parallel to

$$\vec{a} + \vec{b}.$$

Q6. Solve:

$$\tan^{-1} x + \tan^{-1}(x - 1) = \frac{\pi}{4}$$

Q7. Prove that

$$\tan^{-1}\left(\frac{x + y}{1 - xy}\right) = \tan^{-1} x + \tan^{-1} y$$

Q8. If

$$f(x) = \begin{cases} a \sin x + 1, & x < 0 \\ 2, & x = 0 \\ bx - 1, & x > 0 \end{cases}$$

is continuous at $x = 0$, find a and b.

Q9. Find

$$\frac{dy}{dx}$$

if

$$y = \sin^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$$

Q10. Find equations of tangents to

$$y = x^2$$

which are perpendicular to

$$x + 2y = 5$$

Q11. Evaluate:

$$\int \frac{x^2}{1 + 5x^2}\, dx$$

Q12. Find the particular solution of

$$\frac{dy}{dx} \cos x = 1 + y \sin x$$

given that

$$y = 1 \quad \text{when } x = 0$$

Q13. Show that the points

$$(1, 2, -1),\ (2, 1, 3),\ (4, 5, -2),\ (6, 3, -1)$$

are coplanar.

Q14. A bag contains 3 white balls and 2 black balls while another bag contains 2 white balls and 3 black balls. Two balls are drawn and found to be one white and one black. Find the probability that the balls were drawn from the second bag.

Q15. Let

$$a * b = a + b + ab$$

Show that $*$ is commutative and associative.

Q16. Show that

$$y = \frac{4\sin x}{2 + \cos x}$$

is increasing in

$$\left[0, \frac{\pi}{2}\right]$$

Q17. Using integration, find the area of triangle whose vertices are

$$(1, -1),\ (2, -1),\ (1, -2)$$

Q18. If

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix},$$

find

$$A^{-1}$$

— End of Question Paper —